

TITLE

False Green ? Does your monitoring actually monitor?

Chaos engineering for AI pipelines • ROCm-aware

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THE HOOK

"This health check says everything is fine."

Show a green status. Then: it has been returning empty responses for two minutes.

Nothing errored. It reported success the whole time.

THE PROBLEM

Every pipeline is monitored; almost none of the checks can express the failure they exist to catch.

'all clear' means two different things: "no problems found" and "I could not look."

A check that passes while the thing it guards is absent manufactures confidence.

THE FAILURES (this is real, not hypothetical)

Silent CPU fallback ? dashboard green, work 20?40×slower

Empty output ? a guard turns "no data" into a clean '0.000'

Budget exhausted in a hidden reasoning channel ? a result over no data

The monitor never fires ? "zero problems" and "couldn't look" share one output

Truncation at the cap ? reported as success

WHAT FALSE GREEN IS

Chaos engineering aimed at the monitoring itself ? not telemetry analysis.

It injects the failures and asks whether your check would have fired.

A smoke detector tests the smoke detector.

THE VERDICTS

REAL ? the monitor caught the injected failure

GREEN_OVER_NOTHING ? it reported healthy while the failure was live

UNKNOWN ? could not be judged ? never green

THE DEMO (screenshot)

The naive monitor: 5 green-over-nothing

The honest monitor: 5 real

AMD residency panel: four devices, pinned by PCI bus, failing closed.

TECH ? why the AMD stack is load-bearing

ROCm residency read live ('rocm-smi'; the residency signal is the instrument, not a badge)

Pins by PCI bus, never by index; fails closed to UNKNOWN

Runs against a live Ollama-on-ROCm pipeline; deployable to AMD Developer Cloud unchanged

Non-vacuous selftest: remove a rule ? the verdict degrades to UNKNOWN, never passes

Pure Python stdlib core • MIT • one command, no install

VALUE + CLOSE

Every team running inference at scale ships at least one of these green-over-nothing checks and does not know

Converts an invisible cost (wasted GPU-hours, corrupt metrics) into a per-check verdict.

```
'python3 -m falsegreen --selftest'
```

Your pipeline says everything is fine. False Green checks whether it actually looked.

Repo: github.com/Sophia-Thickums/greenover • Demo: sophia-thickums.github.io/greenover